

A State Is Its Relations: The Yoneda Lemma and Relational Identity in International Relations

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Abstract

International Theory's essentialism-trap exchange has left relationalism positioned as both the prescribed exit from fixed-entity essentialism and a candidate for the next essentialism. This paper offers a formal result that threads the bind. Within any category that specifies actors and the relations among them, the Yoneda lemma determines an actor's identity up to indistinguishability by its complete profile of relations to every other actor — and says nothing, in either direction, about the prior choice of category. Three contributions follow. First, a precise formal result: identity is relational profile, scoped to the frame a theorist chooses and to nothing else. Second, a methodological discipline on constructivist identity claims, built by triangulating Abdelal et al.'s content-plus-contestation scheme with Fearon's five referent-senses of state identity and showing that each is a distinct categorical commitment producing a distinct functorial reading of the same actor. Third, a dissolution of the long-standing tension between formal-theoretic and post-positivist IR: the framework asserts closure over nothing, and its reflexive limits — read through Lawvere's fixed-point theorem and Yanofsky's "paradoxes are limitations, not contradictions" frame — become structurally legible rather than destructive. The position is pluralism without relativism: determinate within a frame, undetermined between frames.

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1 Introduction

The essentialism-trap exchange that has run through the pages of *International Theory* over the last three years has produced a working bind: Global IR’s prescribed exit from fixed-entity essentialism runs through relationalism, but a universalized relationalism reinstates essentialism one level up. [Barnett and Zarakol \[2023\]](#) opened the conversation by diagnosing a pattern they call the essentialism trap — “a range of commitments oriented around the notion that the world is constituted by pre-formed, fixed, internally coherent, and bounded social forms” (p. 430) — and prescribed an exit through relationalism and global history. [Bayly \[2023\]](#) answered with a relational sociology of disciplinary knowledge — IR’s intellectual history as “ongoing, provisional, connected, and configurational” (p. 470). [Eun et al. \[2025\]](#) then pushed back: “relationalism is one useful move but not the only one, and . . . its universal imposition would be itself a new form of essentialism (essentialized relationalism)” (p. 275). [Barnett and Zarakol \[2025\]](#) conceded the big-tent point and called for substantive research programs rather than general methodological prescriptions. The exchange leaves relationalism positioned simultaneously as the antidote to essentialism and as a candidate for the next one.

This paper offers a formal result that threads the bind. Within any theoretical frame that specifies its actors and the relations among them, an actor’s identity is determined up to indistinguishability by its complete profile of relations to every other actor in the frame — and the result says nothing, and can say nothing, about the prior question of whether such a frame should be adopted. That asymmetry is the move. It delivers the payoff a relationalist response to essentialism asks for — identity becomes its relational profile, given the frame — while remaining structurally silent on the choice of frame itself. The stance is continuous with [Acharya \[2014\]](#)’s founding account of Global IR as “pluralistic universalism” rather than universalism “applied to all” (p. 652). The result below sits inside that account: it supplies one formal articulation of what relational identity amounts to, conditional on a frame the theorist chooses on grounds the theorem does not adjudicate.

The formal result is the Yoneda lemma, a foundational theorem of category theory [[Mac Lane, 1998](#), [Riehl, 2016](#)]. Once an IR theorist’s object of study — a system of states, an order, a field of practice — is modeled as a category, with actors as objects and their relations as morphisms, the lemma says that each object is determined up to isomorphism by its complete pattern of relations to every other object in the category. In the idiom of the

paper: a state’s identity, within any such category, *is* its relational profile. Crucially, the theorem is conditional. It states what follows from a relational frame; it does not state that IR must be framed relationally, and it does not privilege one relational frame over another. Section 5 proves the lemma and develops its IR content; Section 5.7 states the conditionality explicitly. The first contribution of the paper is thus a formal result: within the scope of any category a theorist is willing to specify, identity is relational profile, no more and no less.

The second contribution is a methodological discipline on constructivist identity claims, built by triangulating the theorem with the two most careful statements the IR literature has produced about what “identity” does and does not mean. Abdelal et al. [2006] answer the question of what identity *is*: content (four non-exclusive types — constitutive norms, social purposes, relational comparisons, cognitive models) plus contestation (the degree to which the content is a matter of internal agreement). Fearon [1999] answers the distinct question of what “state identity” *refers to*: five senses that the literature routinely conflates, from category-membership to historical continuity. The Yoneda framework supplies a structural connection between the two: Abdelal et al.’s content populates the relational profile; Fearon’s five senses become five distinct categorical commitments, each producing a different functorial reading of the same actor. The methodological consequence is that an identity claim must carry its categorical index to be fully specified — a discipline Section 6 develops at length.

The third contribution dissolves what has looked, for four decades, like an irreducible tension between formal-theoretic and post-positivist IR. Critics from Ashley [1988] and Walker [1993] through Tickner [1992], Campbell [1992], Weldes [1999], Epstein [2008], and Zehfuss [2002] have charged that formal frameworks purchase their closure by foreclosing the political, historical, and representational work that makes their objects intelligible in the first place. The charge is correct for frameworks that assert closure over their domain. The theorem presented here does not. It states what follows from a chosen frame and is silent on the choice, and its own reflexive limits — what happens when the framework is asked to describe itself — become, via Yanofsky [2003]’s reading of Lawvere’s fixed-point theorem, structural features rather than contradictions: paradoxes are limitations, not failures. Section 7 develops the engagement with the post-positivist critique in detail, and Section 7.3 shows how the framework’s self-referential limits become productively legible rather than destructively closing.

One paragraph to fix the position the paper is and is not claiming, so it does not have to be re-fixed in every section that follows. The theorem determines identity *within* a chosen category; it is silent on whether to adopt a relational categorical specification at all; and the discipline it implies applies reflexively to the framework itself. The framework is one structural articulation of relational identity among possible others, and the categorical commitment on which the articulation rests — the choice to model actors and relations as a category in the first place — is precisely the kind of theoretical commitment [Eun et al. \[2025\]](#) insist must remain a local move rather than a universal imposition. This is what lets the paper occupy the structural place [Barnett and Zarakol \[2023\]](#) identify for a relationalist response to essentialism without reinstating the trap they warn against, and to sit within [Acharya \[2014\]](#)’s pluralistic universalism rather than on top of it.

Sections 2 through 4 develop the apparatus the theorem requires — categories, functors, natural transformations — in international-relations terms, with running examples drawn from trade and alliance politics. Section 5 states and proves the lemma, works it through the examples, and closes with the conditional-scope statement on which the rest of the paper turns. Section 6 develops the methodological discipline, triangulated with the Abdelal and Fearon literatures and worked through Hopf’s four-discourse reading of Russian identity at the turn of the millennium. Section 7 develops the engagement with post-positivist critique, reads six specific critics through the framework, and invokes Lawvere’s fixed-point theorem to show that the framework’s reflexive limits are legible rather than destructive. Section 8 extends the discussion to frameworks beyond the state-centric default — regional, non-Western, non-state-actor — under the pluralistic-universalism register [Acharya \[2014\]](#) articulates, and Section 9 concludes.

2 What Is a Theory of International Relations?

At minimum, a theory of international relations specifies a domain of actors and a type of relationship between them. Realism foregrounds power among states; liberalism foregrounds institutional and economic ties; constructivism foregrounds the social relationships — recognition, enmity, friendship — through which actors constitute one another.

The discipline’s name is not incidental. International relations takes relations as its subject matter, and its formal tools should treat them as first-class — primitive rather than derivative. Many of the formal tools the field uses go the other way: game-theoretic models take agents, actions, and pay-offs as primitive and treat strategic relationships as consequences; attribute-tabulated statistical work flattens relations into independence-assuming dyadic variables, discarding the compositional structure that connects actors across the system. Category theory takes the other direction: morphisms — the relations the theory posits — are the primitive data,¹ and actors are identified by the pattern of morphisms they participate in, a claim Section 5 will make precise.

A *category* is the mathematical structure formalizing this inversion: a collection of objects with a specified type of relationship (called *morphisms*) between them, subject to minimal axioms any reasonable notion of “relationship” should satisfy. The claim of this section is that a theory of international relations, at the appropriate level of abstraction, is a category.

2.1 Categories as theories

Definition 2.1 (Category). A *category* $\mathcal{C}$ consists of a collection of *objects* $\text{Ob}(\mathcal{C})$; for each pair of objects A, B , a set of *morphisms* $\text{Hom}_{\mathcal{C}}(A, B)$; for each triple A, B, C , a *composition* operation $\circ: \text{Hom}(B, C) \times \text{Hom}(A, B) \rightarrow \text{Hom}(A, C)$; and for each object A , an *identity morphism* $\text{id}_A \in \text{Hom}(A, A)$. These data are required to satisfy associativity ($h \circ (g \circ f) = (h \circ g) \circ f$) and the identity law ($f \circ \text{id}_A = f = \text{id}_B \circ f$).

Each component carries a substantive interpretation. The *objects* are the actors the theory takes as primary — typically states, though nothing in the formalism prevents taking non-state actors, international organizations, firms, or individuals as objects when the theory demands it. The *morphisms* from A to B collect the relationships of the relevant type, typed and directed; the choice of what counts as a morphism is the fundamental theoretical commitment that distinguishes one theory from another. *Composition* says that indirect, mediated connections are themselves relationships of the same type: if A bears a relation to B and B a relation to C , then a composite relation

¹Category theory admits a single-sorted axiomatization in which morphisms are the only primitive and objects are recovered as identity morphisms; the relational ontology can be made the whole formalism, not merely reflected in it.

runs from A to C . The *identity morphism* id_A is the self-relation every actor bears to itself — the domestic economy in a trade category, the commitment to self-defense in an alliance category.² *Associativity* ensures that chaining relationships is unambiguous regardless of grouping.

2.2 The trade category

Example 2.2 (The trade category $\mathcal{T}$). Let the objects of $\mathcal{T}$ be states. For states A and B , a morphism $f \in \text{Hom}_{\mathcal{T}}(A, B)$ represents a trade relationship from A to B — an export flow, a trade agreement, a supply-chain dependency. Composition captures transitive linkage: if $f: A \rightarrow B$ and $g: B \rightarrow C$, then $g \circ f: A \rightarrow C$ represents the indirect trade relationship between A and C mediated by B . The identity morphism id_A represents A 's domestic economy — the self-exchange that leaves other trade relationships unchanged when composed with them.

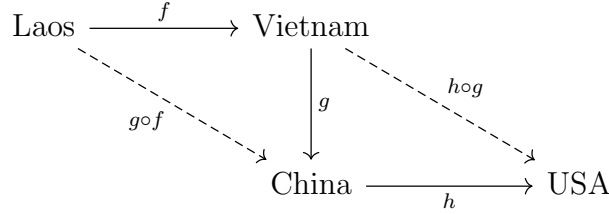


Figure 1: A fragment of the trade category $\mathcal{T}$. Solid arrows are direct trade morphisms; dashed arrows are composites. Associativity guarantees that the fully mediated relationship from Laos to the USA is unambiguous: $h \circ (g \circ f) = (h \circ g) \circ f$.

We will return to this four-state fragment throughout the paper.

2.3 The alliance category

Example 2.3 (The alliance category $\mathcal{A}$). Let the objects of $\mathcal{A}$ be states. A morphism $f \in \text{Hom}_{\mathcal{A}}(A, B)$ represents an alliance commitment from A to B — a mutual defense pact, a security guarantee, an informal alignment.

²The identity *morphism* id_A is distinct from the identity of a state, which is the subject of Section 5 and will turn out to be determined by the totality of A 's relationships to every other actor rather than by the self-loop alone.

Composition captures extended deterrence: if A is allied to B and B to C , the composite $g \circ f$ represents the chain of commitments linking A to C . The identity morphism captures a state's commitment to its own defense.

The alliance category illustrates an important feature of the framework: the same set of states can be organized into different categories by choosing different morphism types. The United States, Japan, and Australia are the same objects in $\mathcal{T}$ and $\mathcal{A}$, but their morphisms — trade flows versus security commitments — are entirely different. The choice of morphisms constitutes a theoretical commitment.

2.4 The choice of category is a commitment

Naming the category — selecting which actors count, which relations are primitive, which compositions cohere — is not a neutral formal preparation for substantive work. It is already the substantive work. A theorist who takes states as objects and trade morphisms as the relations of interest has made a different theoretical and political commitment from one who takes firms, diasporas, or supply-chain nodes as objects, or who takes recognition rather than trade as the relation that matters. Neither choice is adjudicated by the apparatus itself. This is the conditional structure Section 5.7 will state formally, the methodological discipline Section 6 will develop, and the point of contact Section 7 will use to engage the post-positivist critique of formal IR. The framework does not dictate a category; it says what follows from whichever category a theorist is willing to specify.

3 What Is a Model?

A theory, in the sense of Section 2, specifies actors and relations but says nothing about what those relations produce, measure, or mean. The trade category tells us that states are connected by trade, not whether trade enriches, empowers, or endangers them. To interpret a theory — to give it empirical or normative content — we need to assign concrete data to its abstract structure. The mathematical tool for this is the *functor*.

3.1 Functors as structure-preserving maps

Definition 3.1 (Functor). Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A *functor* $F: \mathcal{C} \rightarrow \mathcal{D}$ consists of:

- (i) an assignment of an object $F(A) \in \mathcal{D}$ for each object $A \in \mathcal{C}$;
- (ii) an assignment of a morphism $F(f): F(A) \rightarrow F(B)$ in $\mathcal{D}$ for each morphism $f: A \rightarrow B$ in $\mathcal{C}$;

subject to $F(\text{id}_A) = \text{id}_{F(A)}$ for all objects A and $F(g \circ f) = F(g) \circ F(f)$ for all composable morphisms f, g .

A functor is a structure-preserving map: it sends objects to objects, morphisms to morphisms, and respects both composition and identity.

3.2 Models as functors to **Set**

The general definition allows functors between any two categories, but one target category is of special importance.

Definition 3.2 (Model of a theory). A *model* of a theory $\mathcal{C}$ is a functor $F: \mathcal{C} \rightarrow \mathbf{Set}$, where **Set** is the category whose objects are sets and whose morphisms are functions.

A model assigns to each actor A a set $F(A)$ of associated data and to each relation $f: A \rightarrow B$ a function $F(f): F(A) \rightarrow F(B)$ specifying how that relation transforms the data.³ The functorial axioms carry substantive content: $F(\text{id}_A) = \text{id}_{F(A)}$ says self-relations act trivially on data, and $F(g \circ f) = F(g) \circ F(f)$ says mediated relationships compose coherently — the indirect link from A to C via B yields the same data transformation computed in one step or two. The framing inverts the usual relationship between theory and data: the theory comes first as relational structure, and a dataset is a *consequence* of a choice of model, so different functors on the same theory yield different datasets not because the world has changed but because the interpretive lens has.

³This is Lawvere’s insight: the classical notion of a “model of a theory” in mathematical logic is exactly a functor from the theory (understood as a category) to **Set** [Lawvere, 1963].

3.3 Two models of the trade category

Example 3.3 (Liberal and realist models of trade). Consider two models of the trade category $\mathcal{T}$ from Example 2.2:

- The *liberal model* $F: \mathcal{T} \rightarrow \mathbf{Set}$ assigns to each state A the set $F(A)$ of possible welfare outcomes for A 's economy — real income levels, consumer surplus, gains from specialization. To each trade morphism $f: A \rightarrow B$, it assigns a function $F(f): F(A) \rightarrow F(B)$ tracking how A 's welfare position translates into welfare effects on B through the trade link. The liberal model reads trade as a mechanism of mutual enrichment.
- The *realist model* $G: \mathcal{T} \rightarrow \mathbf{Set}$ assigns to each state A the set $G(A)$ of relative power positions — military capacity, economic leverage, technological advantage relative to competitors. To each trade morphism $f: A \rightarrow B$, it assigns a function $G(f): G(A) \rightarrow G(B)$ that maps A 's power position to the power shift induced in B by the trade relationship. The realist model reads trade as a vector of strategic dependence.

Both F and G are models of the *same* underlying theory $\mathcal{T}$. They agree on who the actors are and what relationships exist between them. They disagree on what those relationships *mean* — on the sets assigned to each state and the functions assigned to each trade link.

This is the precise sense in which the liberal–realist debate about trade is not a disagreement about structure but about interpretation. The two camps share a theory (states linked by trade) and operate with different models (welfare versus power). Whether they can be systematically compared — whether there is a coherent translation from welfare data to power data that respects the trade structure — is a question about *natural transformations*, the subject of Section 4.

4 Translating Between Models

We now have vocabulary for what a theory is (a category) and what a model is (a functor to $\mathbf{Set}$). The liberal and realist models of the trade category interpret the same relational structure in different ways — welfare versus power. Can we systematically translate between them — not just for one

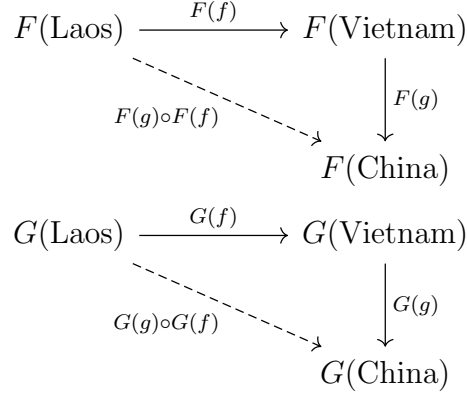


Figure 2: The liberal model F (left) and the realist model G (right) applied to the same fragment of the trade category. Both functors preserve the compositional structure: the indirect link from Laos to China decomposes identically. What differs is the content of the sets and the meaning of the functions.

state, but for every state, and in a way that respects the relational structure? The answer lies in the concept of a *natural transformation*.⁴

4.1 Natural transformations

Definition 4.1 (Natural transformation). Let $F, G: \mathcal{C} \rightarrow \mathbf{Set}$ be two models of a theory $\mathcal{C}$. A *natural transformation* $\eta: F \Rightarrow G$ consists of, for each object A in $\mathcal{C}$, a function $\eta_A: F(A) \rightarrow G(A)$, such that for every morphism $f: A \rightarrow B$ in $\mathcal{C}$, the square $G(f) \circ \eta_A = \eta_B \circ F(f)$ commutes:

$$\begin{array}{ccc}
F(A) & \xrightarrow{\eta_A} & G(A) \\
F(f) \downarrow & & \downarrow G(f) \\
F(B) & \xrightarrow{\eta_B} & G(B)
\end{array}$$

In words: translating A 's data and then pushing it through G 's version of the relationship gives the same result as pushing it through F 's version and then translating B 's data. A natural transformation is not merely a collection of functions chosen state by state; the naturality condition ties those

⁴Eilenberg and Mac Lane introduced categories and functors in 1945 primarily as scaffolding for defining natural transformations [Mac Lane, 1998, p. 29].

functions together across every morphism in the theory. Any translation that works for isolated actors but breaks down when relationships are accounted for is not natural — and, in this framework, not a legitimate translation at all.

4.2 When translation fails: incommensurability

The absence of a natural transformation between two models is not a deficiency of the framework; it is an informative result. It means the two models assign meaning to the relational structure in ways that cannot be coherently reconciled. If no natural transformation $\eta: F \Rightarrow G$ exists between the liberal and realist models of trade, there is no systematic, structure-respecting way to translate welfare data into power data across the international system. The two interpretations are not merely different; they are *incommensurable* in a precise sense.⁵

This gives formal content to a claim that has long circulated in IR metatheory: that paradigms can be genuinely incommensurable, not just different in emphasis. Jackson [2011] argues that different approaches to IR are grounded in distinct philosophical ontologies that resist straightforward comparison; Monteiro and Ruby [2009] push back, arguing that the appearance of incommensurability dissolves under sufficiently careful philosophical analysis. The framework offered here neither settles the philosophical dispute nor requires it to be settled: it reframes the commensurability question as structural. Two models of the same theory are commensurable if and only if a natural transformation exists between them. Whether one does is a mathematical question with a determinate answer once the models are specified, not a matter of interpretation — and Section 7.1 will show that this is precisely how the framework threads the formal-closure / post-positivist-openness tension Section 7 develops.

4.3 The naturality ladder

On the four-state fragment of $\mathcal{T}$ (Figure 1), a natural transformation $\eta: F \Rightarrow G$ must supply translation functions η_A at each state that interlock across

⁵Constant or otherwise trivial families of component maps will often satisfy the naturality squares vacuously; what fails to exist is any η that preserves the relational information on both sides simultaneously.

every trade morphism in the chain. Figure 3 displays the condition as a ladder of commuting squares.

$$\begin{array}{ccc}
F(\text{Laos}) & \xrightarrow{\eta_{\text{L}}} & G(\text{Laos}) \\
\downarrow F(f) & & \downarrow G(f) \\
F(\text{Viet}) & \xrightarrow{\eta_{\text{V}}} & G(\text{Viet}) \\
\downarrow F(g) & & \downarrow G(g) \\
F(\text{China}) & \xrightarrow{\eta_{\text{C}}} & G(\text{China}) \\
\downarrow F(h) & & \downarrow G(h) \\
F(\text{USA}) & \xrightarrow{\eta_{\text{U}}} & G(\text{USA})
\end{array}$$

Figure 3: The naturality condition for $\eta: F \Rightarrow G$ across the four-state trade chain. Each square must commute simultaneously: converting welfare to power state by state is not enough; the conversions must interlock across every trade relationship.

With categories, functors, and natural transformations in hand, we have the first three layers of the framework: a theory, an interpretation of a theory, and a translation between interpretations. The next section isolates the structural fact that makes the whole apparatus cohere — that each actor in a theory is determined, up to indistinguishability, by its complete pattern of morphisms to every other actor.

5 Identity as Relational Profile

So far, every model we have considered has been an external interpretation: a functor that assigns data to the theory’s actors and relations from the outside. The liberal model reads trade as welfare; the realist model reads trade as power. In each case, the interpretive choice — the functor — comes from the theorist.

But there is a family of models that arise canonically from the theory itself, without any external choice at all. These are the *representable functors*, and they encode the idea that a state’s relational profile — the totality of its relationships to every other state — is itself a model. This section introduces the relational profile, states the Yoneda lemma, draws out its consequences, and concludes by making explicit the conditional scope that governs the result and that the subsequent two sections will deploy.

5.1 The relational profile

Definition 5.1 (Representable functor). For any object X in a category $\mathcal{C}$, the *representable functor* $\text{Hom}_{\mathcal{C}}(-, X): \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ is defined by:

- (i) On objects: $\text{Hom}(-, X)$ sends each object A to the set $\text{Hom}(A, X)$ of all morphisms from A to X .
- (ii) On morphisms: given $g: B \rightarrow A$ in $\mathcal{C}$, the function $\text{Hom}(g, X): \text{Hom}(A, X) \rightarrow \text{Hom}(B, X)$ sends $f \mapsto f \circ g$ (precomposition with g).

What does this say? For a fixed state X , the functor $\text{Hom}(-, X)$ assigns to every other state A the set of all relationships that A bears to X . It collects, for the entire system, the answer to the question: “how does the world relate to X ?”

This is X ’s *relational profile* — not an intrinsic property of X , but the complete record of how every other actor in the system is connected to it.⁶

The relational profile is itself a functor to **Set** — a model in the sense of Definition 3.2. But it is a model of a very particular kind. Unlike the liberal or realist models, it makes no interpretive choice about what trade *means*. It simply records, for every state in the system, the full inventory of relationships pointing toward X . It is the most theoretically innocent model available: the model that says “to understand X , look at how the entire world relates to it.”

⁶The notation $\mathcal{C}^{\text{op}}$ indicates the *opposite category*, in which all morphisms are formally reversed. This technical detail ensures that the Hom -functor is covariant (a morphism $g: B \rightarrow A$ in $\mathcal{C}$ induces a function $\text{Hom}(A, X) \rightarrow \text{Hom}(B, X)$ going in the “right” direction). The reader who finds this confusing may safely ignore it; the intuition — $\text{Hom}(-, X)$ collects all relationships pointing toward X — is what matters.

The profile as a working functor. The definition says that $\text{Hom}(-, X)$ acts not only on objects but on morphisms, and this is where the profile acquires its structural force. Consider the trade morphism $g: \text{Vietnam} \rightarrow \text{China}$ in $\mathcal{T}$. The profile $\text{Hom}(-, \text{USA})$ sends g to the function

$$\text{Hom}(g, \text{USA}): \text{Hom}(\text{China}, \text{USA}) \rightarrow \text{Hom}(\text{Vietnam}, \text{USA})$$

that takes each trade channel from China to the USA and extends it backward through g : the direct channel $h: \text{China} \rightarrow \text{USA}$ maps to the composite $h \circ g: \text{Vietnam} \rightarrow \text{USA}$. The profile registers that any trade relationship arriving at the USA from China can be pulled back, via the Vietnam–China link, to a trade relationship arriving from Vietnam. This is precomposition, and it is what makes the profile a functor rather than a mere list: it tracks not just which relationships exist but how they propagate through the network.

5.2 Two profiles in the running example

Example 5.2 (The trade profile of the United States). In the four-state fragment of $\mathcal{T}$ (Figure 1), the representable functor $\text{Hom}_{\mathcal{T}}(-, \text{USA})$ assigns:

$$\begin{aligned} \text{Hom}(\text{Laos}, \text{USA}) &= \{h \circ g \circ f\}, \\ \text{Hom}(\text{Viet}, \text{USA}) &= \{h \circ g\}, \\ \text{Hom}(\text{China}, \text{USA}) &= \{h\}, \\ \text{Hom}(\text{USA}, \text{USA}) &= \{\text{id}_{\text{USA}}\}. \end{aligned}$$

Every state in the fragment has a nonempty set of trade channels pointing toward the USA. The profile is full: the USA is an actor to whom the entire network converges.

Example 5.3 (The trade profile of Laos). The profile $\text{Hom}_{\mathcal{T}}(-, \text{Laos})$ tells a different story:

$$\begin{aligned} \text{Hom}(\text{Laos}, \text{Laos}) &= \{\text{id}_{\text{Laos}}\}, \\ \text{Hom}(\text{Viet}, \text{Laos}) &= \emptyset, \\ \text{Hom}(\text{China}, \text{Laos}) &= \emptyset, \\ \text{Hom}(\text{USA}, \text{Laos}) &= \emptyset. \end{aligned}$$

No other state in the fragment has a trade channel pointing toward Laos. The only relationship Laos bears to itself is the identity — the domestic

economy. The profile is nearly empty: Laos is a source in the trade network, not a sink.

The asymmetry between these two profiles is structural. The USA is an actor to whom many relationships converge; Laos is an actor from whom relationships emanate but to whom few return. The profiles are different because the states *are* different — not in their intrinsic properties (the category says nothing about intrinsic properties) but in how they sit within the relational structure. And the difference is visible at a glance: $\text{Hom}(\text{China}, \text{USA}) = \{h\}$ while $\text{Hom}(\text{China}, \text{Laos}) = \emptyset$.

Example 5.4 (Relational profiles in the alliance category). In the alliance category $\mathcal{A}$, the profile $\text{Hom}_{\mathcal{A}}(-, \text{USA})$ collects all alliance commitments pointing toward the United States: NATO members' commitments, bilateral defense treaties with Japan, South Korea, Australia, the Philippines, and so on. The profile $\text{Hom}_{\mathcal{A}}(-, \text{Switzerland})$ is nearly empty — Switzerland's defining foreign-policy feature is the paucity of alliance morphisms converging on it. In the alliance category, Switzerland and the United States are distinguished precisely by the difference in their relational profiles.

5.3 Reading a profile through a model

A natural transformation $\eta: \text{Hom}(-, X) \Rightarrow F$ is the act of *reading* X 's relational profile through a model F . For each state A , the component $\eta_A: \text{Hom}(A, X) \rightarrow F(A)$ translates every relationship that A bears to X into a datum in F 's language — coherently, because the naturality condition ties the translations together across the relational structure.

The Yoneda lemma says that every such reading is completely determined by a single element: what F assigns to X itself. One datum at X , and the entire interpretation is fixed.

5.4 The Yoneda lemma

Theorem 5.5 (The Yoneda Lemma). *Let $\mathcal{C}$ be a category, $F: \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ a functor, and X an object of $\mathcal{C}$. There is a bijection, natural in both X and F :*

$$\text{Nat}(\text{Hom}_{\mathcal{C}}(-, X), F) \cong F(X).$$

In words: the set of all natural transformations from X 's relational profile into any model F is in exact, one-to-one correspondence with what F assigns to X .

Proof sketch. Given a natural transformation $\eta: \text{Hom}(-, X) \Rightarrow F$, evaluate its component at X on the identity morphism: $\eta_X(\text{id}_X) \in F(X)$. This defines a map $\text{Nat}(\text{Hom}(-, X), F) \rightarrow F(X)$. Conversely, given an element $a \in F(X)$, define $\eta_A(f) = F(f)(a)$ for each $f \in \text{Hom}(A, X)$. Naturality of η follows from functoriality of F , and the two constructions are inverse to each other. $\square$

The proof is short — a few lines of diagram chasing — but the content is deep. The identity morphism id_X acts as a “universal probe”: every natural transformation out of $\text{Hom}(-, X)$ is completely determined by what it does to id_X . This is the formal expression of the idea that X 's self-relation, combined with the relational structure, pins down everything.

What the bijection means. Every element $a \in F(X)$ is not a local fact about X but a *global structural commitment*: the rule $\eta_A^a(f) = F(f)(a)$ propagates a outward through every relationship converging on X . To say “the USA has welfare level a ” is a statement about the *entire trade network as seen from the USA's position*. China's welfare assessment is not a separate parameter; it is $F(h)(a)$, forced by a and the liberal model's transmission function. Vietnam's is $F(g)(F(h)(a))$; Laos's is $F(f)(F(g)(F(h)(a)))$. Each is a consequence of the single datum a , propagated through the relational structure. There are no free parameters left.

Conversely, every coherent reading of X 's relational profile through F is generated by some element of $F(X)$, and no two elements generate the same reading. The local datum and the global interpretation are the same object, viewed from different ends.

5.5 The Yoneda lemma at work

Let us see the bijection concretely in the four-state trade category, taking $X = \text{USA}$.

The liberal model. Take F to be the liberal model of Example 3.3. A natural transformation $\eta: \text{Hom}(-, \text{USA}) \Rightarrow F$ must assign, for each state A ,

a function $\eta_A: \text{Hom}(A, \text{USA}) \rightarrow F(A)$ that is compatible with trade morphisms. The Yoneda lemma says all such transformations are determined by a single element: $\eta_{\text{USA}}(\text{id}_{\text{USA}}) \in F(\text{USA})$.

Pick any welfare outcome $a \in F(\text{USA})$ — say, a particular level of consumer welfare. This determines a unique natural transformation η^a by the rule:

$$\eta_A^a(f) = F(f)(a) \quad \text{for each } f \in \text{Hom}(A, \text{USA}).$$

Tracing the consequences:

- $\eta_{\text{China}}^a(h) = F(h)(a)$: the welfare effect on China of its direct trade link to the USA, starting from the USA's welfare level a .
- $\eta_{\text{Viet}}^a(h \circ g) = F(g)(F(h)(a))$: the welfare effect on Vietnam, obtained by first tracing a through China and then through Vietnam.
- $\eta_{\text{Laos}}^a(h \circ g \circ f) = F(f)(F(g)(F(h)(a)))$: the welfare effect on Laos, at the end of the fully mediated chain.

A single welfare datum for the USA generates a complete welfare distribution across the system. A different datum $a' \neq a$ generates a different distribution — different at every state where the transmission functions $F(h)$, $F(g)$, $F(f)$ are not constant. The bijection $\text{Nat}(\text{Hom}(-, \text{USA}), F) \cong F(\text{USA})$ is now visible: the USA's welfare level is not an independent assignment but a compressed summary of the entire system's welfare structure, as seen from the USA's position.

The realist model. The same bijection applies to the realist model G : each power position $b \in G(\text{USA})$ propagates through the trade network via G 's transmission functions, distributing power assessments where F distributes welfare — from the same relational profile, through different lenses.

Why the profile of Laos works differently. The Yoneda lemma applies equally to Laos, but the bijection looks very different. Since $\text{Hom}(\text{Viet}, \text{Laos}) = \text{Hom}(\text{China}, \text{Laos}) = \text{Hom}(\text{USA}, \text{Laos}) = \emptyset$, a natural transformation $\eta: \text{Hom}(-, \text{Laos}) \Rightarrow F$ has nothing to assign at Vietnam, China, or the USA — the empty set admits only the empty function. The entire natural transformation is determined by $\eta_{\text{Laos}}(\text{id}_{\text{Laos}}) \in F(\text{Laos})$, but it carries no information to any other state. The liberal model's welfare assessment of Laos does not propagate

through the system, because no trade channels converge on Laos through which consequences could propagate.

This is the formal expression of a structural fact: in this fragment of the trade network, Laos is a peripheral actor whose relational profile gives models little to work with. The USA’s welfare level is a global commitment; Laos’s welfare level, in this fragment, is a local island. The difference is not in the states’ intrinsic properties but in the richness of their relational profiles — and the Yoneda lemma makes the difference precise.⁷

5.6 The Yoneda embedding

The Yoneda lemma has an immediate and powerful corollary.

Corollary 5.6 (The Yoneda Embedding). *The functor $\mathbf{y}: \mathcal{C} \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ defined by $X \mapsto \text{Hom}(-, X)$ is fully faithful. That is, for any objects X and Y in $\mathcal{C}$:*

$$\text{Hom}_{\mathcal{C}}(X, Y) \cong \text{Nat}(\text{Hom}(-, X), \text{Hom}(-, Y)).$$

In particular, $X \cong Y$ if and only if $\text{Hom}(-, X) \cong \text{Hom}(-, Y)$.

This is the punchline. Two states are isomorphic — structurally identical within the theory — if and only if they have identical relational profiles. There is no identity beyond relations.

The embedding in the running example. The profiles computed in Examples 5.2 and 5.3 make this concrete. The USA and Laos cannot be isomorphic in $\mathcal{T}$, because their profiles disagree: $\text{Hom}(\text{China}, \text{USA}) = \{h\}$ while $\text{Hom}(\text{China}, \text{Laos}) = \emptyset$. A natural isomorphism $\text{Hom}(-, \text{USA}) \cong \text{Hom}(-, \text{Laos})$ would require, at the component for China, a bijection between a one-element set and the empty set — which does not exist. The Yoneda embedding thus distinguishes the USA from Laos on purely relational grounds: not because they differ in GDP or population or regime type, but because the morphisms converging on them are different.

⁷The structural significance of cardinality variation in $F(X)$ — of which the empty-function case for Laos is one limit — is developed in Section 6.3 and its discussion of ontological security and erasure.

5.7 The conditional scope of the theorem

The Yoneda embedding establishes that identity is relational profile — within a chosen category. This conditionality is not a stylistic hedge appended to the theorem; it is the form of the theorem. The proofs and constructions of this section all take place inside a category we have already chosen: trade morphisms and alliance commitments as primitives, the particular four states of the running example as objects, specific compositional axioms as structural constraints. The theorem determines what identity follows from that choice. The theorem does not, and cannot from within its own resources, adjudicate the choice itself.

Changing the category changes the relational profile. The USA’s profile in the trade category (Example 5.2) is dense with morphism-structure; its profile in the alliance category (Example 5.4) is organized around entirely different morphism types, with a substantially different set of non-empty profile entries. Each profile is a legitimate identity-assignment within its own category; neither is the USA’s “true” identity. The theorem is available on equal terms to rationalists, constructivists, institutionalists, and critical theorists who may adopt very different categorical commitments; what it delivers, given any such commitment, is a structural statement about what identity within that commitment looks like.

The identity the theorem assigns is therefore perspectival, with the perspective fixed by the categorical commitment and the perspectives disciplined by the naturality condition that Section 4 introduced. Section 6 develops the methodological consequence that follows for IR practice: an identity claim is formally ill-formed unless it specifies the model through which the claim is made. Section 7 pursues the conditional scope in another direction, showing that it is the framework’s specific response to post-positivist concerns about the costs of formal closure, and that — as Section 7.3 will establish via Lawvere’s fixed-point theorem — no framework rich enough to describe itself can coherently claim a scope any less conditional than this one.

6 Identity Is Perspectival: The Methodological Move

The perspectivalism of the preceding section has a methodological consequence before it has a metaphysical one: an identity claim that does not

carry its model is not a cautious claim but a formally ill-formed one, because the theorem says nothing about identity except within a chosen category. This section states the discipline that follows from this observation, shows that it is consonant with two canonical IR-methodological literatures on identity, works it through a concrete empirical case, and closes by grounding the discipline in the theorem rather than in stylistic preference. The framework developed here is one structural articulation of relational identity among possible others. Adopting a relational categorical specification is itself a theoretical commitment the theorem does not adjudicate from outside, and the discipline applies reflexively: our own invocation of the Yoneda framework is a categorical choice the discipline obliges us to carry.

6.1 The discipline: identity claims must specify a model

Consider the claim “Russia perceives NATO as enemy.” The claim is intelligible in ordinary language and routinely deployed in IR prose; within the framework of Section 5, however, it is ill-formed. Identity, including the hostile-perception relation between two actors, is determined only within a chosen category: a specification of which actors count, which relations are primitive, and which compositional moves cohere. Stated through a trade-volume category, the claim contracts to near-vacuity, since trade morphisms between Russia and NATO members are dense and largely cooperative, and a hostile-perception reading of that morphism structure would require the category to contain resources it does not contain. Stated through an alliance-commitment category, the claim is nearly tautologous, since the morphism structure is built precisely to distinguish committed partners from unaligned or opposed states. Stated through a historical-memory discourse category of the kind Hopf [2002] constructs, the claim becomes substantive: it picks out a specific discursive positioning within a landscape of competing identity-readings. These three readings are not rival empirical claims about Russia; they are projections, through different functorial models, of three different categorical commitments.

The discipline is not a demand for methodological monism. Different categorical commitments are appropriate for different questions, and the framework is agnostic about which is right in any given case. What the discipline demands is that an identity claim carry its index — that the model through which the claim is made be specified, so that the profile is determinate and the claim’s force and scope are legible to readers who may work in different

commitments. Section 8 returns to what this implies for pluralism without relativism; Section 7.1 restates the conditional form of the theorem that generates the discipline.

6.2 Triangulation with existing identity methodology

The discipline is neither an external norm imposed on IR nor a restatement of what the field already does; it is structurally consonant with two methodological literatures on identity that have been developed independently and used side-by-side without theoretical unification. The framework’s contribution is to supply the structural glue.

Abdelal, Herrera, Johnston, and McDermott on identity as a variable. Abdelal et al. [2006] argue that the social-scientific study of identity has been plagued by conceptual looseness, and propose to rescue identity as an analytical variable by decomposing it along two dimensions. The first is content: collective identity, they argue, is populated by some combination of four non-mutually-exclusive types — constitutive norms, social purposes, relational comparisons, and cognitive models.⁸ The second is contestation: identities are not fixed within groups, but vary in the degree to which members agree on their content. Three of the four content types are explicitly relational. Constitutive norms are morphisms to normative positions in the category (whatever it takes to be a member of some identity-relevant group); social purposes are morphisms to goal-objects or outcome-objects; relational comparisons are morphisms to other identity groups. The fourth type, cognitive models, is the one least obviously fitted to a relational reading, and for good reason: cognitive models index *which* morphism-types and which objects are recognized as identity-relevant in the first place, which is a commitment at the level of category-choice rather than of morphism-content. Contestation, on this reading, is the empirical observation that the profile a model extracts from an actor is underdetermined until a particular categorical commitment and its morphism-assignments are fixed — the gap between a category’s morphism structure and any particular functor that realizes it.

⁸For the typology see Abdelal et al. [2006, pp. 695–697]; constitutive norms “refer to the formal and informal rules that define group membership,” social purposes “refer to the goals that are shared by members of a group,” relational comparisons “[define] an identity group by what it is not,” and cognitive models “refer to the worldviews or understandings of political and material conditions and interests that are shaped by a particular identity.”

Fearon on the five referent-senses of “state identity.” Fearon [1999], working in analytic-philosophical mode, observes that “state identity” in IR usage refers to at least five distinct things that the literature routinely conflates: membership in a category of states (major power, democracy, NATO member), self-respect features that distinguish one state from others (historical memory, symbolic markers), the sovereignty-literature sense of what it means to be a state at all, continuity through time in the sense philosophers of personal identity debate, and the frequent conflation of state identity with national identity.⁹ Each of Fearon’s senses is, in the framework’s vocabulary, a distinct categorical commitment. The first is a commitment to a category whose objects include grouping-objects (memberships) and whose morphisms include membership relations from states into them. The second is a commitment to a category whose objects include historical events and symbolic markers, with self-reference morphisms linking a state to the memories it carries. The third is a meta-level commitment concerning which objects count as states in the first place — what the sovereignty literature calls the institutional content of stateness. The fourth is a diachronic-category commitment, requiring more apparatus than Section 5 supplies, though it fits naturally within indexed-category or fibred-category extensions.¹⁰ The fifth is a category-selection error: conflating objects that live in two different categories, with no functor between them making the conflation coherent. Fearon’s taxonomy, read structurally, ceases to be a list of what “state identity” might mean and becomes a map of category-choices the IR literature makes when it deploys the phrase.

The structural glue. Abdelal et al. and Fearon are not in conversation, and their methodological anchors have been used independently. The framework supplies the unification. Abdelal et al. answer the question *what populates an identity profile*; Fearon answers the question *which category the profile lives in*; the Yoneda lemma answers the question *what identity is, given those two specifications*. The three answers compose: once a reader has specified a Fearon-sense (which category), an Abdelal-content (which morphism-types populate the profile), and a functor (which interpretive model extracts

⁹See Fearon [1999, §8] for the disentanglement; the manuscript circulates widely but remains unpublished.

¹⁰The continuity-through-time case is flagged in Section 9 as a natural direction for future development rather than treated here.

data from the profile), the framework determines what identity-claim the reader is making. The claim is not that either Abdelal et al. or Fearon would endorse this categorical translation of their work; it is that the translation is available and coherent, and that it renders the discipline of specifying-the-model a structural consequence of resources already present in the identity-methodology literature.

6.3 The structural range of a model-specification

A model-specification is formally a functor $F: \mathcal{C} \rightarrow \mathbf{Set}$; for each actor X , the model assigns a set $F(X)$. The cardinality and structure of $F(X)$ are themselves structural data about the actor, and two limit cases are diagnostically important. When $F(X) = \emptyset$, the actor is erased within the model. Erasure propagates backward: any actor A with a morphism into X has its image under F forced to $\emptyset$, because no function from a nonempty set to the empty set exists. Model-level erasure is therefore never local; it reorganizes everything the model can say about actors who relate to the erased one. This is the structural home of several of the critiques developed in Section 7.2: Tickner [1992]’s diagnosis that IR categories systematically erase household, reproductive, and care-based relations is, in structural terms, the observation that defaulted functors send large classes of actors to $\emptyset$; Ashley [1988]’s heroic-practice critique exposes the same erasure at the level of sovereign-authority categorization; and Campbell [1992]’s performative account of foreign-policy constitution is, structurally, an account of the representational work required to lift $F(X)$ from near-empty to nonempty for any particular other. When $F(X) = \{*\}$ is a singleton, the actor is present but structurally inert within the model: every incoming morphism collapses to the unique function into the one-element set, and every outgoing morphism is a point-selection from $F(B)$, making the actor into a forced-choice participant without independent content. Ontological security studies calls variation in the richness of $F(X)$ for an actor’s own self-model *degrees of ontological (in)security* [Mitzen, 2006], and this framework supplies the specific cardinality claim behind that variation: full ontological security is a rich and compositionally stable $F(X)$, reactive NPC-status is $F(X) = \{*\}$, and ontological erasure is $F(X) = \emptyset$. Naming these as structural homes is not a claim that the critical literature needed categorical translation to be coherent. The claim is only that the framework is expressive enough to register the moves these literatures make, rather than filtering them out of the formal

apparatus.

6.4 A worked case

Weldes and the security imaginary as category choice. Weldes [1999]’s analysis of the 1962 Cuban Missile Crisis turns on the observation that the Soviet missile deployment was not self-evidently a threat to US national interests; making it a threat required substantial representational work within what she calls the postwar US *security imaginary*. “The world of international politics, the place of a particular state within it, and the consequent national interests attributed to that state are not objective facts to be discovered; instead, they are the products of contingent, historically forged articulations of linguistic and symbolic elements already present within the security imaginary of the state” [Weldes, 1999, p. 10].¹¹ Weldes’s security imaginary is a category choice made concrete. It fixes which actors count (states, alliances, transnational communist movements, domestic publics); it fixes which relations are primitive (threat-transmission, ideological affinity, sphere-of-influence claims); it fixes which compositions cohere (the domino theory is a compositional axiom; the Munich analogy is a morphism-type that links analogous situations). The framework does not import foreign vocabulary into Weldes’s discursive-constructivist analysis; it supplies the structural statement of the move her work already makes, and in doing so, renders the category-choice aspect of her argument formally visible rather than tacit.

Hopf on the four identity discourses of 1999 Moscow. The framework’s sharpest vindication comes from a case where multiple categorical commitments are visible and active side-by-side within a single empirical setting. Hopf [2002]’s study of Russian identity and foreign policy in 1999 provides exactly this: “With the collapse of the Soviet Union, Russia found itself between two different modern identities — that of the Soviet past and that of the western present. Four different discourses characterized Russia’s position with regard to these two modernities” [Hopf, 2002, p. 153]. The four are New Western Russian, New Soviet Russian, Liberal Essentialist,

¹¹Elsewhere: “in the process of articulation such extant linguistic resources are combined and recombined to produce contingent and contextually specific representations of the world. The security imaginary furnishes the rules according to which these articulations are forged” [Weldes, 1999, p. 10].

and Liberal Relativist. Each, read structurally, is a distinct categorical commitment. The New Western Russian discourse commits to a category whose objects are Western-order institutions and whose morphisms are integration-and-legitimation relations; Russian identity under this functor is the profile of how Russia relates to, and is received by, the institutional West. The New Soviet Russian discourse commits to a category organized around the Historical Other of the Soviet Union, with civilizational-bloc and Slavic-affinity morphisms that the Western-order category does not carry. The Liberal Essentialist discourse commits to a category in which Russia, the West, and the East are elements that together synthesize into a Russian identity; the morphisms include civilizational-contribution relations not reducible to either bloc alone. The Liberal Relativist discourse commits to a category with a reflexive structure: identity-claims are themselves objects, and the characteristic morphism is the refusal to fix any particular identity as final — a self-referential structure that Section 7.3 will connect to Lawvere’s fixed-point result, since self-reference of this kind is precisely what that result bounds. NATO’s 1999 bombing of Serbia becomes an especially clean test of the framework, because Hopf records the same event being read through all four categorical commitments. Under the New Soviet Russian discourse, “the two Slavic brothers enjoyed a natural affinity, cemented still more by sharing the danger of NATO’s assault from the air” [Hopf, 2002, p. 228].¹² Under the Liberal Essentialist discourse, “Russia, comprised of different tribes, does not defend Serbs, but rather the right of the peoples of Yugoslavia to freedom and independence, including Kosovan Albanians” [Hopf, 2002, p. 234]. Under the New Western Russian discourse, the bombing is an institutional-sovereignty problem — the West’s departure from the legal-normative regime Russia hopes to join. Under the Liberal Relativist discourse, the bombing is largely absent from foreign-policy implication, which is itself the structural signature of a category whose reflexive refusal to fix identity projects resists commitment to any foreign-policy stance. One event, four profiles, four identity-readings of Russia. The framework’s additive contribution is not to replace Hopf’s discursive analysis but to supply a question his vocabulary cannot pose: whether natural transformations between these four functorial models exist, and, where they do not, what structural obstructions make the

¹²Page references to Hopf’s Chapter 5; the NATO–Serbia case runs across several sections of that chapter and this draft gives representative rather than comprehensive citations.

four discourses formally incommensurable rather than merely different. The classification is of course based on our exegetic reading of Hopf, not on any claim he makes.

Katzenstein and the reach across constructivist lineages. The framework’s utility is not specific to Russia or to discourse analysis. Katzenstein [1996] explains Japan’s post-1945 non-militarization through norms — both regulative and constitutive — that are treated as social facts with causal force rather than as discursive representations, a methodological commitment substantially different from Hopf’s. Within the framework, Katzenstein’s analysis corresponds to a different category: objects include norm-institutions (police institutions, self-defense force practices, legal constraints on military action); morphisms are norm-effect relations (compliance, internalization, causal shaping of behavior); and the compositional axioms are those of social-fact causation rather than of discursive articulation. A functor on this category yields a Japanese identity-profile that is not merely a different content-assignment from Hopf’s Russian profiles but a profile in a structurally different category, accommodated by the framework without collapsing the methodological distance between the two constructivist lineages.

6.5 Back to the theorem

The discipline — specify the model through which an identity claim is made — is not a stylistic preference or an external methodological imposition. It is the direct methodological consequence of the conditional form in which the Yoneda theorem establishes the relation between identity and relational profile in Section 5. The theorem determines, for any chosen category and any functor on it, what identity the theorem assigns to each actor. What the theorem does not determine, and what it structurally cannot determine from within its own resources, is which category to choose or which functor to adopt. This asymmetry — determinate within a chosen commitment, undetermined between commitments — is what the framework contributes to the identity-methodology literature that Abdelal et al. and Fearon have mapped and that Weldes, Hopf, and Katzenstein have worked in concrete cases. It is also what makes the theorem a meaningful tool for the post-positivist critics of formal closure, rather than a target for their critique. The discipline, then, is a theorem-level consequence rather than a stylistic imposition; what remains is to show that the same conditionality that generates the discipline

is what lets the theorem meet post-positivist critics on the ground they have spent four decades preparing, to which Section 7 turns.

7 Closure, Openness, and the Limits of Formalism

7.1 The theorem is conditional

Section 5 proved that within any category $\mathcal{C}$, an object's identity is determined up to isomorphism by its relational profile — the totality of morphisms every other object bears to it. This is the Yoneda lemma, stated and proved inside the categorical apparatus. It is important to state what the theorem does and does not claim. The theorem claims that *given* a chosen category, identity takes a specific relational form: every datum a model assigns to an actor is a projection of that actor's relational profile, and two actors are structurally indistinguishable iff their profiles agree. The theorem does not claim that any particular category is the right one, that categories are mathematically privileged over non-categorical ways of thinking about international politics, or that adopting a relational framework for identity is a conclusion the mathematics delivers. The choice of category — which actors, which typed relations, which compositional axioms — is a theoretical commitment that precedes the theorem and constrains its reach. The theorem says what follows from the commitment; the commitment is not adjudicated from inside the theorem.

This conditional form is not a qualification added for diplomacy. It is the form of the result itself. A reader who extracts an unconditional structural claim from the framework — that identity is relational ontology *tout court*, or that the Yoneda lemma vindicates relationalism against its critics — is reading the theorem as something it is not. The theorem is a structural statement about the internal consequences of any categorical commitment, available on equal terms to theorists who adopt very different commitments: realist and constructivist, rationalist and reflexivist, mainstream and critical. What it buys is the ability to say, cleanly and within any given commitment, what counts as the identity of an actor. What it does not buy, and does not try to buy, is a settlement of the question of which commitment to adopt. This conditionality is what makes serious engagement with post-positivist critics possible in the following subsection. If the theorem asserted

unconditional closure, those critics would be right to reject it on exactly the grounds they have rejected unconditional closure claims for forty years. Because the theorem does not assert unconditional closure, the engagement is substantive rather than adversarial.

7.2 Engagement with post-positivism

Post-positivist scholars have spent four decades diagnosing the costs of formal closure in international relations theory. The Yoneda theorem, stated conditionally as in Section 7.1, does not settle that diagnosis; it takes it seriously by refusing the move those critics target. This section works through six distinct post-positivist concerns — the structural, the performative, the gendered, the deconstructive, the applied, and the internal-constructivist — and shows that the framework’s conditional scope is responsive to each. The point is not that the theorem vindicates relational ontology against its critics. The point is the reverse: the critics have been right that unconditional structural-closure claims are untenable, and the conditional-scope framing is how the theorem honors that finding rather than resisting it.

Walker: the inside/outside dialectic as historically produced. [Walker \[1993\]](#) reads modern IR as “a discourse that systematically reifies an historically specific spatial ontology, a sharp delineation of here and there” (p. ix): a set of claims whose self-evidence depends on treating the inside/outside distinction as given when it is in fact historically specific and politically produced. Formal models that proceed as if sovereignty were settled participate in that reification. Our framework makes the cut explicit rather than dimming it. A category choice *is* the inside/outside cut in Walker’s sense — a determination of which actors are in, which relations matter, which compositional moves cohere — and the framework requires that determination to be stated, not assumed. The theorem says what follows from a given cut; it does not claim the cut is necessary, essential, or final. Walker’s critique is therefore not an objection to the framework but a reading of what the framework makes explicit: the choice of category is contestable, and the theorem’s force extends no further than the choice does.

Campbell: foreign policy as performative constitution. [Campbell \[1992\]](#) presses the performative version of this critique. Foreign policy, in

his reading, is not a response to pre-given threats but a discursive practice that produces the identity of the state by inscribing its outside; danger is interpreted into existence through representational work that casts specific differences as alien, subversive, or threatening. Without that work, the inside/outside cut does not stay cut. This pushes Walker's point one step further: the categorical commitment is not a static structural reification but an ongoing discursive labor, and each snapshot of a chosen category is the momentary output of a practice whose maintenance requires continuous representational work. The Yoneda theorem holds within any given snapshot. What Campbell's critique shows is that the framework cannot settle the question of which snapshot is the right one; that question is answered, contestedly, by the practices Campbell traces.

Tickner: the binary oppositions are gendered. [Tickner \[1992\]](#) argues that the binary oppositions on which IR has been built — inside/outside, order/anarchy, public/private, reason/emotion, autonomy/relatedness — are systematically gendered: the first term of each pair is associated with masculinity, the second with femininity, and the hierarchical arrangement privileges masculine-coded relations in ways the discipline has rarely interrogated. The implication for a framework like ours is specific. The choice of which category to work in is not only politically produced in Walker's sense or performatively maintained in Campbell's sense; it is gendered. Realist and liberal categorical framings foreground states, alliances, power, and rational calculation — all masculine-coded. Categorical commitments that foreground household-to-state morphisms, care-to-security morphisms, reproductive-to-productive morphisms, or kinship-to-authority morphisms would yield different relational profiles, different naturalness constraints, different identity-determinations. The framework does not generate these alternative categories on its own; that generative work is substantive feminist IR's, and the framework cannot substitute for it. What the framework does is refuse the claim that the defaulted categories are neutral. Every theorem holds within a category; no category is privileged by the theorem.

Ashley: the heroic practice and the refusal of deep structure. [Ashley \[1988\]](#) offers the most technically demanding of these critiques and the one most directly aimed at formal work. His target is the "heroic practice" of modern IR theory: a hierarchical dichotomy in which sovereignty is

treated as a foundational rational presence from which anarchy is derived as its necessary external domain. Formal models are Ashley’s canonical example: their deductive power depends on taking sovereignty as a settled foundation from which structural consequences follow. Read uncharitably, the Yoneda framework fits this description — axiomatic categories, theorems deducing identity-determinations, an apparatus treating sovereignty as prior to relations. The framework’s conditional scope is the specific refusal of that reading. The theorem does not take the category as foundational presence; it takes the category as an explicit, contingent, contestable commitment and says what follows within it. Ashley’s formulation of the state as “an intrinsically contested, always ambiguous, never completed construct — a construct that is itself always in the process of being imposed in the face of never-quieted resistances” (p. 231) is, in our language, the claim that no categorical commitment fixes the state once and for all. The framework makes that claim structural: a different category yields a different profile; no category is adjudicated from outside.

Weldes and Epstein: applied post-positivism. [Weldes \[1999\]](#) and [Epstein \[2008\]](#) show the preceding critiques operating on concrete empirical terrain. Weldes’s treatment of the Cuban Missile Crisis and her “security imaginary” concept are examined at length in Section 6.4, where the security imaginary is read as the category choice made concrete. [Epstein \[2008\]](#) traces the mid-twentieth-century shift from a pro-whaling to an anti-whaling international order, showing it cannot be explained by material interests but was produced by the emergence of a powerful discourse that restructured state identities as pro- or anti-whaling positions. In our framework, the anti-whaling discourse is again a category choice, and the resulting state positions are profiles within it. Neither Weldes nor Epstein uses formal machinery; both treat identity as discursive-constitutive. The framework does not replace their analyses but supplies the structural statement of what the case work demonstrates: once a security imaginary or discourse has fixed the category, the Yoneda theorem determines the resulting identity-profiles, and the interesting analytical question shifts from “what is Russia’s identity?” to “through which imaginary is Russia being read?”

Zehfuss: the internal critique. The most demanding engagement is with [Zehfuss \[2002\]](#), whose argument is not an external critique of construc-

tivism from a post-positivist outside but an internal critique from within the Wendtian tradition our paper extends. Zehfuss’s core move (ch. 5, “The impossibility of pure presence”) is that constructivism depends on a notion of reality as raw material from which construction proceeds, and that this notion is incoherent on Derridean grounds: raw material is always already interpreted, and constructivism’s attempt to stabilize its interpretations as being *about* a pre-given reality reproduces the positivist problems it claimed to overcome. Her chapter 2 shows that Wendt’s “identity move” — making identity the key constructivist concept while treating collective identities as things that can change, stabilize, and be stabilized about — cannot be sustained on Wendt’s own terms, because each attempt to fix identity as object of theory reinstalls the pure-presence move the tradition claims to refuse.¹³ The Yoneda framework is directly exposed to this critique. If we claimed, unconditionally, that identity *is* relational profile, we would be performing exactly the stabilization move Zehfuss diagnoses: treating relational content as raw material from which identity facts are extracted. The conditional-scope framing is the specific response. The theorem does not claim unconditional identity-is-relational-profile; it claims conditional identity-is-relational-profile-given-a-chosen-category, and the chosen category is itself a commitment the framework does not stabilize. What Zehfuss calls the impossibility of pure presence is, in our language, the structural fact that no category is privileged by the framework and no profile is unconditioned by the choice. Her critique is, read carefully, what our paper builds on rather than resists: formalization that refuses the conditional scope commits the stabilization move she attacks; formalization that foregrounds the conditional scope does not.

Across these six engagements, the pattern is consistent: the post-positivist tradition has been right that unconditional structural-closure claims about identity are untenable, and the framework’s conditional scope accepts that finding rather than resisting it. What remains is a narrower claim — a statement about the internal structure of any categorical commitment, available to critics and proponents of that commitment on equal terms. Whether this narrower statement earns the name “theorem” without reintroducing the closure its critics have targeted is itself a structural question, with a structural answer supplied by the next subsection.

¹³The book-length argument was prefigured in [Zehfuss \[2001\]](#).

7.3 Incompleteness and openness

There is a final structural result to invoke, one which converts the conditional scope from a methodological stance into a categorical theorem. Consider what happens when the Yoneda framework is applied reflexively to its own apparatus. If we construct a category whose objects are themselves categorical commitments — the trade category, the alliance category, Hopf’s New Western Russian discourse, Weldes’s postwar security imaginary — and ask the theorem to determine an identity for each by its relational profile to the others, we are asking the framework to describe itself. For any sufficiently rich categorical framework that can formally internalize its own model-structure in this way, self-reference has structural limits, and those limits are what makes the conditional scope of Section 7.1 not merely a methodological preference but a structural necessity.

Lawvere [1969] established the limit in its canonical form: in any Cartesian closed category, if an object Y admits any endomorphism without a fixed point, then no weakly point-surjective morphism $A \rightarrow Y^A$ can exist in the category for any A .¹⁴ The force for a self-describing framework is precisely this: the structural preconditions for totalizing self-reference — the ability to represent all maps into a function-space Y^A as points of the domain A , where Y carries even minimally expressive structure — cannot be satisfied without contradicting the presence of fixed-point-free maps that any expressive category inherently contains. Self-reference of the relevant kind is structurally bounded in any rich categorical framework, and no appeal to mathematical care at the edges of the framework will lift that bound.

Yanofsky [2003] summarizes the upshot in a single sentence: “paradoxes are limitations, not contradictions.” The fixed-point obstructions Lawvere’s theorem characterizes do not break a framework that encounters them; they mark its inherent structural limits, and those limits sit exactly at the points where totalizing identity claims would generate inconsistency. The framework’s inability to totalize over its own apparatus is not a failure to be repaired but the structural feature that keeps it coherent as a conditional-scope tool. Read in this direction, the post-positivist insistence that unconditional closure is impossible is not merely philosophically defensible but categorically

¹⁴The result, now known as Lawvere’s fixed-point theorem, is the categorical form behind Cantor’s theorem, Russell’s paradox, Tarski’s undefinability of truth, and Gödel’s first incompleteness theorem. See Yanofsky [2003] for an accessible exposition of the theorem and its disparate instances.

guaranteed. The framework agrees with Walker, Campbell, Tickner, Ashley, Weldes, Epstein, and Zehfuss not only as a matter of interpretive charity but as a matter of theorem: any framework expressive enough to describe itself must carry structural limits of exactly the form their critiques describe.

The engagement in Section 7.2 met six distinct post-positivist concerns about closure; this subsection supplies the structural foundation on which those engagements rest. The conditional scope established in Section 7.1, from which Section 6’s methodological discipline follows, is not a diplomatic courtesy extended to critics but the only scope a self-describing framework can coherently claim. The closure post-positivists have for forty years been right to worry about is not merely philosophically suspect; it is, for any framework rich enough to describe itself, mathematically unavailable — and the conditional scope that generates the discipline of Section 6 is the structural form a framework of this kind must take to remain coherent.

8 Comparing Frameworks and Pluralism without Relativism

The preceding sections developed the framework within a single theory: a category $\mathcal{C}$, its models, translations between models, and the Yoneda lemma that pins identity to relational profile. But international relations does not consist of a single theory. The trade category, the alliance category, and a recognition category are three different theories of the same system of states. They share the same actors but differ in the relations they foreground. How do we compare them?

8.1 Functors between theories

The same tool that gives us models — the functor — also gives us theory translations.

Definition 8.1 (Theory translation). A *theory translation* from $\mathcal{C}$ to $\mathcal{D}$ is a functor $P: \mathcal{C} \rightarrow \mathcal{D}$: an assignment of objects to objects and morphisms to morphisms that preserves composition and identity.

The definition is identical to Definition 3.1, but the conceptual shift is important. When the target is **Set**, a functor interprets a theory in the

world of data. When the target is another theory, a functor translates one relational vocabulary into another.

Example 8.2 (From trade to alliance). Suppose major trade partnerships tend to generate security commitments — that states which trade heavily develop overlapping interests that eventually crystallize into alliance obligations. This hypothesis can be expressed as a functor $P: \mathcal{T} \rightarrow \mathcal{A}$ that maps each state to itself (the actors are the same) and each trade morphism to the alliance commitment it is hypothesized to induce. The functorial axioms impose constraints: if the trade links from Laos to Vietnam and from Vietnam to China together generate a composite alliance commitment from Laos to China, then P must map the composite trade morphism $g \circ f$ to $P(g) \circ P(f)$ — the induced commitment must respect the chain of relationships, not just the bilateral links. Not every hypothesized connection between theories will satisfy these constraints, and the failure is informative: it says the two theories’ relational structures are incompatible in a specific way.

8.2 Pulling back models

Theory translations do more than relate relational structures. They transport models. If $P: \mathcal{C} \rightarrow \mathcal{D}$ is a functor between theories and $F: \mathcal{D} \rightarrow \mathbf{Set}$ is a model of $\mathcal{D}$, then the composite $F \circ P: \mathcal{C} \rightarrow \mathbf{Set}$ is a model of $\mathcal{C}$. It assigns to each object of $\mathcal{C}$ whatever F assigns to its image in $\mathcal{D}$, and to each morphism the corresponding function. The composite is called the *pullback* of F along P .

This is how cross-theory interpretation works. A conflict model that assigns conflict probabilities to states and conflict-transmission functions to relations in a conflict category can be pulled back along a functor from the trade category to the conflict category: the result is a model of trade that reads trade as conflict, assigning to each state the conflict probability induced by its trade position and to each trade morphism the corresponding conflict-transmission function. Different functors P encode different causal hypotheses; whether a natural transformation exists between the pullback models of competing hypotheses is a determinate structural question of the kind Section 4 developed.

8.3 Reducibility and embedding

The properties of a theory translation $P: \mathcal{C} \rightarrow \mathcal{D}$ characterize the structural relationship between the two theories. P is *faithful* if it preserves distinctions: whenever $\mathcal{C}$ has two different morphisms $f \neq g$ between the same pair of objects, $P(f) \neq P(g)$ in $\mathcal{D}$ — the target can express every relational distinction the source makes. P is *full* if, conversely, the source captures every relation in the target: for any morphism $h: P(A) \rightarrow P(B)$ in $\mathcal{D}$, there exists $f: A \rightarrow B$ in $\mathcal{C}$ with $P(f) = h$. P is *fully faithful* if both conditions hold; this is an *embedding*, with the source as a sub-theory of the target, no relational information lost or gained.

These distinctions bear on long-standing debates about theory reduction in IR. Can the alliance category be reduced to the trade category — do all alliance commitments arise from trade relationships? This is the claim that a full functor $P: \mathcal{T} \rightarrow \mathcal{A}$ exists. Can trade relationships be distinguished within the alliance framework — do alliance commitments preserve the fine-grained structure of trade? This is the claim that P is faithful. The answers are empirical, but the framework makes the structural content of the claims precise: a reducibility claim is a claim about a specific functor with specific properties, not a vague assertion that “everything is really about trade” or that “alliances subsume trade.”

8.4 Pluralism without relativism

The apparatus now delivers a specific metatheoretical position. Different categories are different theories of the same world; the Yoneda lemma determines identity rigorously within each, but the framework supplies no mathematical procedure for choosing among them. Where a functor between two theories exists, the theories are comparable — reducibly, embeddably, or partially — and the precise sense of comparison is readable off the functor’s properties. Where no functor exists that respects both theories’ relational structures, the theories are structurally incommensurable in the sense of Section 4.2, not because one is incoherent but because they organize the same actors around relations that do not translate. This is the formal counterpart of Jackson [2011]’s pluralism: determinate within each framework, undetermined between frameworks, and neither relativist (no “anything goes” — each framework carries its own rigorous internal structure) nor absolutist (no privileged category from which the others are derivative).

The framework’s reach extends beyond the state-centric default of the paper’s running examples. Different categorical commitments yield different kinds of actors: states, regional institutions, transnational networks, practice communities, firms. [Acharya \[2001\]](#)’s ASEAN case is a category in which the primary objects include a regional institution alongside its member states, with morphisms tracking the dense web of consultation, informal consensus, and norm-diffusion practices that constitute the security community — a categorical commitment quite different from the state-centric realist default, and one in which the Yoneda lemma’s identity claim applies to ASEAN as an actor with its own relational profile. [Keck and Sikkink \[1998\]](#)’s transnational advocacy networks live in a different categorical commitment again — objects are NGOs, intergovernmental bodies, foundations, and states; morphisms are information flows, boomerang pressure patterns, and framing relationships. In each case the theorem holds: actors within the chosen category are determined by their relational profiles. The choice of which category to work in is a theoretical and political commitment the framework does not adjudicate, which is exactly the formal expression of [Acharya \[2014\]](#)’s principle that Global IR “subsumes, rather than supplants, existing IR theories and methods” (p. 654).

9 Discussion and Conclusion

This paper asked what follows, formally, from the commitment to a relational ontology in international relations — and insisted from the first page that the commitment itself remains outside the theorem’s scope. The answer: within any such commitment, a state’s identity is nothing more and nothing less than its complete relational profile, the totality of relationships every other actor in the system bears to it.

The argument proceeded in layers. A theory of international relations is a category: a collection of actors and typed relations, with composition and identity (Section 2). A model is a functor to **Set**: a structure-preserving interpretation that assigns data to actors and functions to relations (Section 3). A translation between models is a natural transformation: a systematic conversion of one interpretation into another that respects the relational structure at every point (Section 4). The Yoneda lemma (Section 5) proved that every datum a model assigns to a state is a global structural commitment — a complete reading of the state’s relational position, compressed into

a single element — and the Yoneda embedding proved that two states are structurally identical if and only if their relational profiles agree.

The three contributions announced at the outset are consequences of that chain. The *formal result* is the theorem itself, scoped by the chosen category and by nothing else. The *methodological discipline* is the requirement that an identity claim carry its categorical index (Section 6), which triangulates the Abdelal content–contestation scheme with the Fearon typology of referent-senses into a single structural move: identity claims become claims about a relational profile in a specified category, neither more determinate than the category permits nor less determinate than the theorem guarantees. The *dissolution of the formal-closure / post-positivist-openness tension* (Section 7) follows from the conditional scope itself: the framework asserts closure over nothing, and its reflexive limits — Lawvere’s fixed-point theorem read through Yanofsky’s “paradoxes are limitations” frame — make the self-referential boundaries structurally legible rather than destructively closing.

The framework has limitations. It is static, not dynamic: categories describe configurations, not trajectories, and a proper account of identity in motion would need at least a 2-categorical treatment. It says nothing directly about agency or intentionality; those require enrichment of the morphism structure with commitments the present paper does not make. It makes identity theory-dependent — a feature for constructivists, a cost for positivists seeking theory-independent foundations, and for us a theorem-level consequence rather than an evasion of either position. Extensions to enriched categories, indexed and fibred categories, higher categories, sheaves, and dependent type theory are natural next steps for addressing these limitations, and each is already developed within category theory for comparable purposes in other domains.

The core result stands without qualification, where the qualification is precisely the point. In any theory a scholar is willing to specify as actors, typed relations, composition, and identity, the Yoneda lemma guarantees that identity is relational. No hidden essence, no intrinsic property, no attribute not already encoded in the relational profile can distinguish two actors whose profiles agree. A state is its relations — within the frame the theorist chooses to work in, and not beyond it.

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